

Bron transformasie

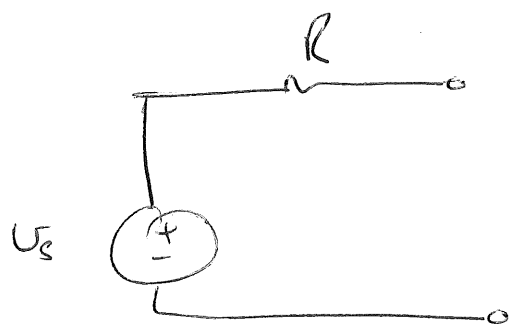
→ Ekwivalente kringbane:

- * Kringbaan A is ekwivalent aan kringbaan B indien beide dieselfde V-I karakteristieke het.

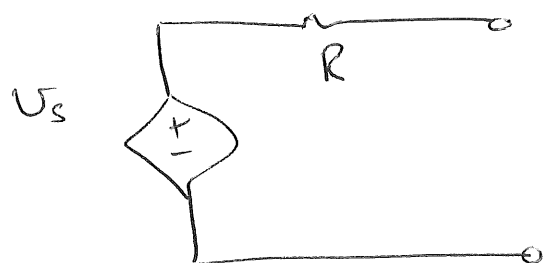
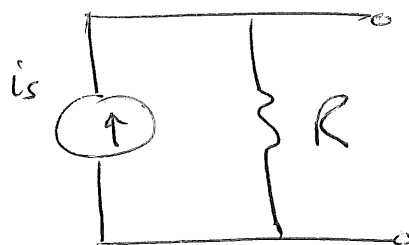


→ Bron transformasie:

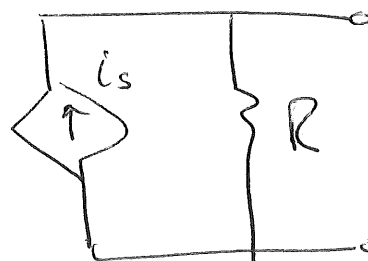
- * Vervang V-bron en 'n series-R met 'n I-bron en 'n parallel-R.



$$\longleftrightarrow U_s = i_s R$$



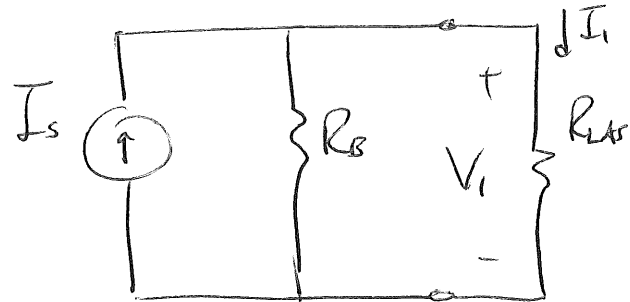
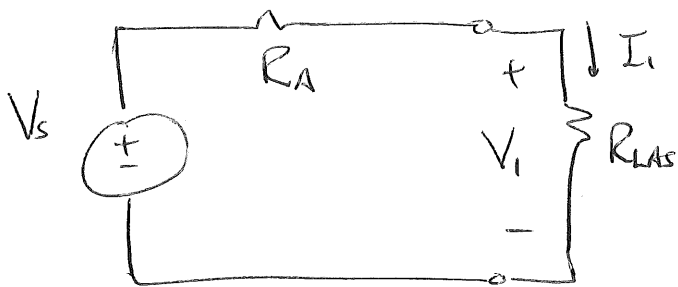
$$\longleftrightarrow U_s = i_s R$$



* Perspectief van de las.

(2)

* Die las R_{LAS} behoort nie die verskied agter te kom nie, its spanning en stroom, indien dit aan kringbaan A of B gekoppel is nie.



$$V_i = \frac{V_s}{R_A + R_{LAS}} \times R_{LAS}$$

$$V_i = I_s \times \frac{R_B R_{LAS}}{R_B + R_{LAS}}$$

$$I_i = \frac{V_s}{R_A + R_{LAS}}$$

$$I_i = I_s \frac{R_B}{R_B + R_{LAS}}$$

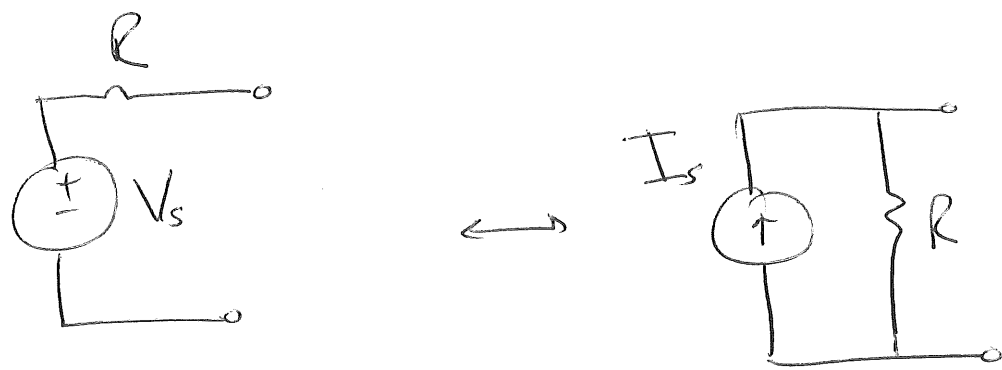
→ Vir ekwivalente kringbane:

$$\frac{V_s}{R_A + R_{LAS}} \times R_{LAS} = I_s \times \frac{R_B R_{LAS}}{R_B + R_{LAS}}$$

$$\frac{V_s}{R_A + R_{LAS}} = I_s \frac{R_B}{R_B + R_{LAS}}$$

→ Stel $R_A = R_B$. Dus:

$$\rightarrow V_s = I_s \cdot R_A = I_s \cdot R_B$$



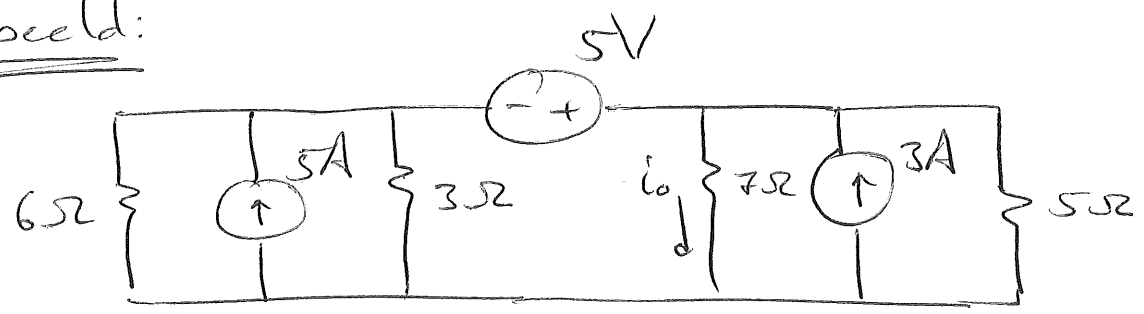
* $V_s = I_s R$

* Weerstand is gelyk.



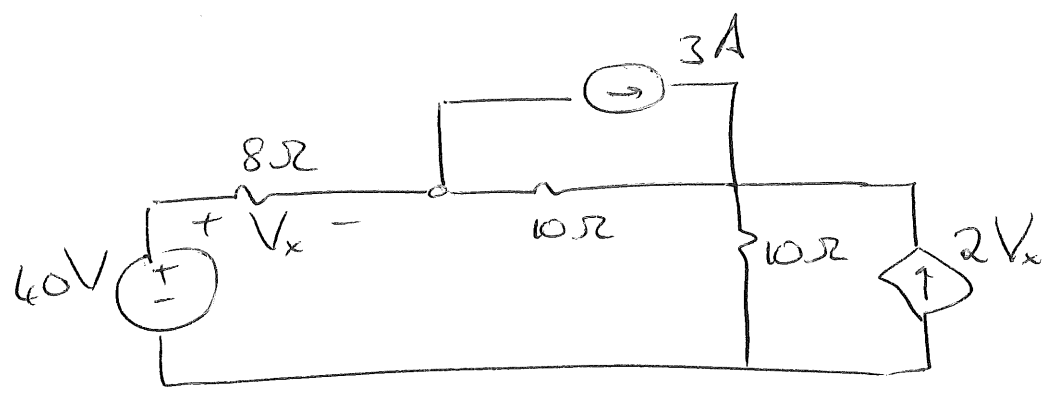
Voorbeeld:

①



$i_o = ?$

②

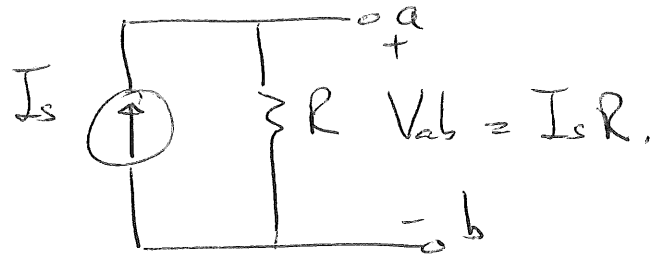
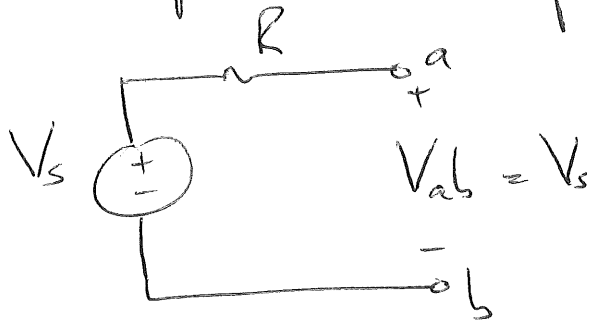


$V_x = ?$

Addisionele analyses:

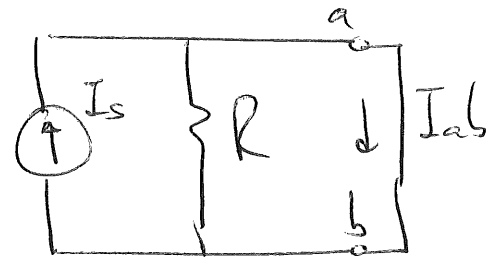
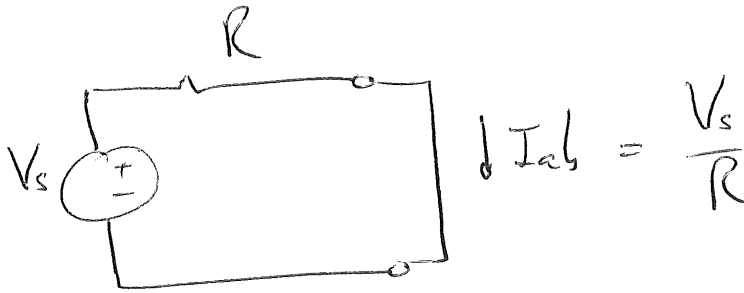
(4)

1 * Ope-baan spanning $\rightarrow V_{ab}$



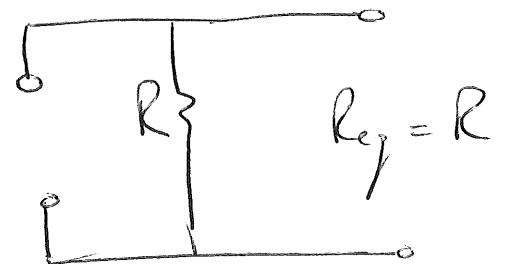
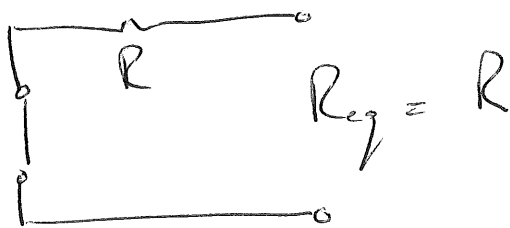
Equivalent indien $V_s = I_s R$.

2 * Kortsluit stroom $\rightarrow I_{ab}$



Equivalent indien $\frac{V_s}{R} = I_s$, $I_{ab} = I_s$.

3 * Equivalente weerstand.
(Stelk bronne et.)



$\rightarrow$ Kringbare is equivalent indien

- + ope-baan spanning dezelfde is, en
- + kortsluit stroom dezelfde is, en
- + equivalente weerstand dezelfde is.